

SEAN MACBRIDE

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EDUCATION

University of Zürich, Zürich, Switzerland

May 2024 - Present

Ph.D. in Physics

Thesis title: *Standard Siren Cosmology: From Instrumentation to Inference*

University of Michigan, Ann Arbor Michigan

Aug 2022 - May 2024

M.Sc. in Physics

Wheaton College, Norton Massachusetts

Aug 2016 - May 2020

B.A. in Physics with Honors. Minor in Astronomy

Thesis title: *Characterizing the Dust and Cold-Gas Content of Nearby Star-Forming Galaxies.*

RESEARCH

University of Zürich

Apr 2023 - Present

Ph.D Student

Prof. Marcelle Soares-Santos

- **Rubin Observatory: LSST-Camera (LSSTCam) Summit Support Scientist, Target-of-Opportunity (ToO) Coordinator.**
 - Led verification, integration, and testing of LSSTCam to the Rubin Observatory on Cerro-Pachón in Chile from receipt on summit through science verification and early operations.
 - Characterized defects in LSSTCam, including the picture-frame response in different detector types and vampire pixels. Developed custom image processing task for monitoring stability of defects in LSSTCam focal plane.
 - Led the ToO observing program during the Rubin science verification period (May 2025 - December 2025), including ToO responses to gravitational wave (GW) candidates S250725j and S251112cm.
- **Dark Energy Science Collaboration (DESC):** Co-Founder of the Standard Sirens Topical Team, Collaboration Council Representative (2026-2028).
 - Led a study of galaxy catalog completeness on dark siren analysis consequences. Obtained a 1.4M CPU-hour computing grant from the National Energy Research Scientific Computing Center to support this research. Expected publication mid 2026.
 - Preparing a dark-siren analysis using LSST commissioning data (Data-Preview 2) and public data from the LIGO-Virgo-KAGRA collaboration, the first dark siren measurement using LSST. Results in early 2027.
- **Dark Energy Survey (DES):** Dark Energy Camera (DECam) Image Processing and Analysis, Partnership with Subaru Scientists.
 - Implemented improvements to Dark Energy Survey gravitational wave (DESGW) alert and image processing pipeline, including improvements to the telescope strategy and real-time alert monitoring.
 - Performed observations, image processing, and analysis of images from the DECam in response to GW triggers from International Gravitational-Wave Observatory Network (IGWN) observing run 4 (O4).
 - Organized collaboration between the Japanese collaboration for Gravitational wave ElectroMagnetic follow-up (J-GEM) and DESGW through joint observations of GW event S250328ae using DECam and the Prime Focus Spectrograph (PFS) at Subaru Observatory.

- **IGWN:** H_0 estimation with the DECam Local Volume Exploration survey (DELVE) and DES, ToO coordination with Rubin Observatory.
 - Led dark siren analysis using DELVE survey data for IGWN O4c data, including galaxy catalog preparation, developing analysis pipelines for IGWN O4c analysis, and systematic-uncertainty quantification.
 - Coordinated time-critical ToO observations with IGWN researchers, interfacing with Rubin Observatory alert brokers to rapidly disseminate information about GW candidates.
 - Contributed to dark siren analysis using DES survey data for IGWN O4b data.
- **Image Sensors:** Prototype Low-Noise Image Sensor Testing and Optimization
 - Designed a new testing apparatus for testing image sensors for astronomical applications using vacuum, thermal, and optical subsystems.
 - Optimize readout noise in Oscura CCDs through testing different clock and multi-sampler configurations.
- **Robotic Positioners:** Testing and Optimization for Multiplexed Spectroscopic Surveys
 - Tested Dark Energy Spectroscopic Instrument (DESI) robotic positioners to increase movement precision and improve low performing robotic positioners on the DESI focal plane.
 - Characterized a common failure mode of DESI positioners and resolved the failure with a reproducible solution. Authored and submitted an internal report and presented the results of this study to the DESI focal plane working group.
 - Led dynamic testing of prototype robotic positioner performance through lifetime, focal-ratio degradation, and fiber angle tests using different telescope configurations. Tested prototype robotic positioners for stage V surveys, focusing on characterization of tilt induced by different orientations.

University of Michigan & Southwest Research Institute

Feb 2023

Graduate Observer - NASA LUCY

Prof. David Gerdes (UM) & Marc Buie (SwRI)

- Participated in a ground-based observation campaign to study the occultation of the Jupiter trojan asteroid 15094 Polymele. Authored procedure used by 50+ team members, detailing process to configure Celestron telescope mount, optics, and software in efficient manner.
- Coordinated 1,100+ miles of travel to observation site location. Trained other team members in telescope setup and resolved issues with hardware and software in preparation for occultation event.
- Successfully captured field of observation during occultation event. Contributed data to NASA-LUCY ground based occultation team, which confirmed 15094 Polymele's surface features and presence of smaller orbiting satellite.

Massachusetts Institute of Technology

Nov 2020 - Jul 2022

Technician - LLAMAS

Gábor Fűrész (PI) & Mark Egan (Engineering supervisor)

- Assembled and modified opto-mechanical mounts and ground support equipment for Large Lenslet Array Magellan Spectrograph (LLAMAS) instrument according to assembly drawings. Inspected custom-fabricated parts using a coordinate measurement machine and prepared reports detailing measurements.
- Tested the efficiency of diffraction gratings at nominal orientation. Modified existing test equipment to measure blaze-angle transmission. Authored report detailing procedure, methods, results, and analysis.
- Prepared adhesives for bonding handling tabs to optical components. Bonded camera lenses into bezels. Developed and modified bonding procedure to best accommodate changing circumstances.
- Designed protective covers, fixtures, and tools using SolidWorks. Collaborated using different version control software - Git and SolidWorks PDM. Wrote software for precise control of DC servo motors, used to ensure highest standards of spectrograph fiber bonding and integration.

- Loaded and unloaded optical fibers into anti-reflective coating fixtures. Integrated the AR-coated fibers to the spectrograph by bonding with optical adhesive, with 100% accuracy. Authored and modified the fiber bonding and protective tube-pulling procedure to meet evolving science and safety needs.

Massachusetts Institute of Technology

May 2021 - Nov 2021

Research Assistant - STARSPOOT

Gábor Fűrész (MIT) & Jennifer Burt (NASA-JPL)

- Developed a data pipeline for concatenating single-day observations from a multi-channel optical solar spectrometer. Modified pipeline routine to maintain compatibility with different data structures. Utilized data pipeline to collect spectrophotometry from different solar events.
- Created data analysis tools and objects for studying solar events obtained by spectrometer. Compared ground-based data to International Space Station observations from same epochs. Studied correlations between historical S-index, magnetic activity, total solar irradiance, and spectral solar irradiance.
- Determined limit of detection of solar events for the ground-based optical spectrometer. Pipeline and data tools served as supporting evidence in several forthcoming publications that describe the proposed space-based project scope.

University College London & Wheaton College

May 2019 - Jul 2020

Honors Thesis Student

Prof. Amélie Saintonge (UCL) & Prof. Dipankar Maitra (WC)

- Coordinated joint thesis project between UCL and Wheaton College. Authored honors thesis detailing scientific rationale, results, and central conclusions of study. Presented to Wheaton and UCL faculty and students for critique and discussion. Received highest marks from faculty of both schools.
- Designed and implemented python data pipeline for use with derived data from xCOLD GASS, JINGLE, and SDSS galaxy surveys. Developed a linear Markov-chain Monte Carlo sampler to constrain relationship of cold-gas and dust components of star-forming galaxies from Balmer emission. Generalized the sampler to higher dimensions to include the inclination-dependent reddening in the calibration.

University College London

Jan 2019 - Mar 2019

Group Lead

Prof. Thanh Nguyen

- Led research group of 8 UCL students to determine the detection limits of lateral flow assays using photo-thermal heating of magnetic nanoparticles. Resolved conflicts between group members and determined group structure. Coordinated laboratory logistics and safety measures between student group and the Royal Institute of London.
- Prepared ferromagnetic nanoparticle solution and samples for observation. Designed setup of infrared laser and sample apparatus. Obtained and analyzed temperature data from a thermal camera for determination of optimal membrane type and solution concentration.
- Collaborated and authored report detailing group processes, results, & analysis. Received highest marks from UCL faculty and Royal Institute staff.

Rutgers University New Brunswick

May 2018 - Aug 2018

REU Research Assistant

Prof. Carlton Pryor

- Improved proficiencies in Pandas, NumPy, SciPy, and other data analysis tools to support investigations of dwarf satellite galaxy orbits.
- Designed, implemented, and modified data pipeline for use with ESA Gaia space telescope DR-2 data selections of dwarf satellite galaxies. Discovered small scale tidal tail around central core of dwarf satellite galaxy Carina using radial density measurements.
- Prepared and presented progress reports to Prof. Pryor on a weekly basis. Prepared and delivered final presentation at summer research symposium and to professors and graduate students of the physics and astronomy department.

Wheaton College
Student astronomy projects

Aug 2017 - Dec 2018
Prof. Dipankar Maitra

- Designed and 3D printed a mount for projection of sunlight on a 4.5" reflector telescope. Obtained and analyzed solar spectrum using a spectroscope. Observed spectral limb-reddening resulting from gas densities similar to the chromosphere. Authored report detailing methods, results, and analysis.
- Retrieved archived photometry from V404 Cygni outburst event obtained at Wheaton College in June 2015. Analyzed data using Aperture Photometry Tool. Identified two outburst events on the night of June 27, separated by roughly one hour. Authored report detailing methods, results, and analysis.
- Determined specifications of diffraction grating for use with DSLR camera. Obtained spectrum of Vega and Capella using grating and 30s exposure. Identified iron absorption features in both stars. Authored report detailing methods, results, and analysis.
- Retrieved x-ray binary data taken by MAXI x-ray telescope. Implemented Lomb-Scargle periodogram techniques to derive orbital periods of four different x-ray binary systems. Derived slight differences in period of binary 2S 1417-624. Authored report detailing methods, results, and analysis.

Project P.A.N.O.P.T.E.S.
Student Collaborator

Sep 2017 - Dec 2018
James Synge (Google Cambridge)

- Communicated between professors at Wheaton and outside collaborators to install, operate, and maintain autonomous exoplanet telescope inside Wheaton College observatory dome.
- Aligned P.A.N.O.P.T.E.S. exoplanet telescope for observation according to polar-alignment procedure using hand tools.
- Performed maintenance on the astronomy dome to ensure continuous observations using P.A.N.O.P.T.E.S.
- Modified existing dome control code for weather-automated and manual remote operation of telescope.

NASA Langley Research Center
Summer Intern

May 2017 - Aug 2017
Dr. Brian Walsh

- Designed, built, and operated a 2.1 & 1.06 μm pump source for proof of concept atmospheric-monitoring mid-infrared Lanthanide-LuAG laser.
- Programmed a model to determine resonator stability based on resonator dimensions and elements. Determined resonator quality through spectroscopic tests.
- Performed spectroscopy on resonator components using mid-infrared and optical spectrometers. Optimized resonator setup from outcomes of spectroscopic tests and model predictions.

Wheaton College
Undergraduate Research Assistant

Dec 2016 - May 2017
Prof. John Collins

- Built Nd:YAG laser resonators from individual components. Modified existing laser configurations as necessary, including soldering new trigger circuits for a q-switch.
- Determined threshold energies for spontaneous emission using different resonator specifications.
- Exercised safe and appropriate use of voltmeters, oscilloscopes, and other electric and optical equipment.

TEACHING

University of Michigan Physics Department

Lead Graduate Student Instructor

May 2023 - Jun 2024

Ann Arbor, MI

- Organized course administration, including worksheet development, lab procedures, and grading practices, for introductory physics lab that serves 1000+ students and ~15 graduate instructors each term.
- Acted as a liaison between undergraduate students, graduate instructors, introductory lab support staff, and faculty. Resolved grading disputes through collecting all pertinent information, meeting with undergraduate students, and meeting with department chairs and parents of students, on occasion.
- Led three teaching workshops for first-year students over four days throughout the academic year. Reformed training workshops for new graduate student instructors to better prepare them for teaching. Created additional workshops with other lead instructors and new graduate instructors to familiarize new instructors with how to manage classroom social dynamics.

University of Michigan Physics Department

Graduate Student Instructor

Aug 2022 - Dec 2022

Ann Arbor, MI

- Led laboratory sections of undergraduate students in life sciences disciplines through weekly labs focused on introductory physics concepts.
- Student feedback average of 4.72/5 in teacher evaluations related to instruction. Answered questions pertaining to lab content, fundamental concepts, and course policies throughout lab session.
- Held office hours once a week to assist students in all introductory physics classes with homework problems, exam preparation, and comprehension of fundamental concepts of physics. Organized meetings with struggling students outside of usual hours and tailored class sessions to better meet students needs

Wheaton College Physics Department

Teaching Assistant

Aug 2018 - Dec 2019

Norton, MA

- Increased engagement of Introductory Physics I & II students with in-class problems & labs through effective communication and classroom instruction.
- Performed laboratory setup and breakdown for class of 40+ students in accordance with schedule.
- Communicated student comprehension of specific topics to professors to increase participation.

Wheaton College Physics Department

Physics Tutor

Aug 2018 - Dec 2018

Norton, MA

- Assisted students with understanding concepts of physics to support problem sets and exam preparation.
- Increased engagement with struggling students by meeting outside of regular tutoring hours, communicated deficiencies to professors to optimize the classwork plan.
- Participated in tutor development meetings to enhance instruction and communication skills.

OUTREACH

Rubin Observatory

LSST Summit-Support Scientist

Jan 2025 - July 2025

Cerro Pachón, Chile

- Provided interviews to the BBC, the Wall Street Journal, Science News, and other publications regarding LSSTCam.
- Answered questions from journalists over email regarding LSSTCam statistics, commissioning status, and other information in line with DOE/NSF embargo policies.

- Gave tours to media and local dignitaries to showcase LSSTCam performance, features, and known issues.

University of Michigan-Southern Illinois University
Eclipse Group Leader

Aug 2023 - May 2024
Albuquerque NM, Burlington VT

- Led high school group of students to operate a solar telescope and take images of the sun during the 2023 annular eclipse and 2024 total eclipse.
- Teach high students from about physics of the sun, solar observations, and eclipses through virtual and in person presentations.
- Successfully operated and captured images of the 2023 annular eclipse on site in Albuquerque NM, and 2024 total eclipse on site in Burlington VT.
- Encouraged safe and responsible observations and discussed the science of eclipses with the public through several local news outlets, including NBC-5, Vermont Public Radio, and the Essex Reporter

Wheaton College Astronomy Department
Observatory Guide

Aug 2016 - May 2020
Norton, MA

- Showcased features of different telescopes to local tour groups at Wheaton College Observatory.
- Operate and adjust telescopes to show appropriate stars, objects, and events based on sky conditions.
- Educated children and adults in how to locate objects and explained features about the objects on sky.

MENTORSHIP

University of Zürich

- Banan Yamani: Previous Masters Student. Current PhD student at Paul Scherrer Institute (PSI)
- Yavuz Gencel: Previous Masters Student.

University of Michigan, Ann-Arbor

- Elise Kesler: Previous undergraduate student. Physics PhD student at CU Boulder.
- Tim Fanning: Previous undergraduate student. Physics PhD student at Indiana University Bloomington.

PUBLICATIONS

Below are selected publications from different collaborations and projects. A full list of my publications can be found on my Google Scholar profile or on the astrophysics data system.

Astronomical Instrumentation

1. The On-Sky Performance of the LSST Camera CCD Array. **MacBride**, Utsumi, ... in prep.
2. First year into operations of LLAMAS, the Large Lenslet Array Magellan Spectrograph: design, build and commissioning overview, results and performance, lessons learned. Furesz, ... **MacBride**, ... in prep.
3. Skipper CCD readout time optimization for astronomical applications. Wüthrich, **MacBride**, ... in prep.
4. LSST Camera Electro-Optical Test (Run 7) Results. The LSST Camera group: Antilogus, ... **MacBride**, ... 5 May 2025.

5. The Stage-5 Spectroscopic Experiment. Besuner, Dey, ... **MacBride** ... 12 Mar 2025.

Cosmology

1. Dark Sirens in the LSST Era: Evaluating Biases and Analysis Choices. **MacBride**, Andrade-Oliveira, ... in prep.
2. Measurement of the Hubble constant using the Dark Energy Survey Year 6 Gold galaxy catalog and the fourth Gravitational-Wave Transient Catalog. McMahon, ... **MacBride**, ... in prep.
3. Standard Sirens in 2040s: Probing the Cosmic Expansion History with Gravitational Waves and Spectroscopic Galaxy Surveys. Borghi, ... **MacBride**, ... 20 Dec 2025.

Time Domain Astrophysics

1. The Rubin Observatory Mock-Data Challenge. **MacBride**, Chatterjee, ... in prep.
2. Searches for Electromagnetic Counterparts of Gravitational Wave Events during LIGO-Virgo-KAGRA Observing Run 4 using the Dark Energy Camera. McMahon, **MacBride**, ... in prep.
3. A Joint Search for the Electromagnetic Counterpart to the Gravitational-Wave Binary Black-Hole Merger Candidate S250328ae with the Dark Energy Camera and the Prime Focus Spectrograph. Zhang, Kokubo, **MacBride**, ... 1 Aug 2025.
4. NSF-DOE Vera C. Rubin Observatory Observations of Interstellar Comet 3I/ATLAS (C/2025 N1) . Chandler, Bernardelli, ... **MacBride**, ... 21 July 2025.
5. Rubin ToO 2024: Envisioning the Vera C. Rubin Observatory LSST Target of Opportunity program. Andreoni, Margutti, ... **MacBride** ... 7 Nov 2024.

Astronomical Systems

1. The Rubin Observatory Target-of-Opportunity Program in the First Year of Operations. **MacBride**, Ribeiro, ... in prep.
2. LSSTCam installation and commissioning. Fanning, **MacBride**, ... in prep.
3. Scheduling the LSST: commissioning the Rubin Feature Based Scheduler. Jones, ... **MacBride**, ... in prep.
4. The Implementation of a Robust Observing Strategy for Discovering Optical Counterparts to Gravitational Wave Events. Fuller, **MacBride**, ... in prep.

DEPARTMENT SERVICE

- Helped to organize the Conferences for Undergraduate Women in Physics (CUWiP) 2024 at the University of Michigan-Ann Arbor.
- Designed art installation in University of Michigan Physics help room showcasing previous experiences by graduates of the undergraduate program, highlighting struggles overcame and success in their later lives.
- Member of University of Michigan physics graduate council from 2022 - 2024.
- Lead organizer and host of physics graduate student symposium in 2023, a weekly speaker series highlighting research projects from several departments in the University of Michigan.
- Member of UM physics DEI committee from 2022 - 2024.
- Member of APS chapter at UM from 2022 - 2024.

➤ Member and peer advisor of the Wheaton College physics club from 2016-2020.

PRESENTATIONS

Talks

Entries marked with * indicate an invited talk.

- 11/2025* Status of the Rubin Target of Opportunity Program, bi-annual meeting of Rubin-LSST France at the Laboratory of Subatomic Physics & Cosmology, Grenoble, France
- 09/2025* Status of the Rubin Target of Opportunity Program, 5th Philip Wetton Workshop, Oxford University, England
- 07/2025* LSSTCam Commissioning Status, Rubin Community Workshop 2025, Tucson AZ, USA
- 06/2025* Status of the Rubin Target of Opportunity Program, LIGO Seminar Series, Virtual
- 06/2025* Status of the Rubin Target of Opportunity Program, MIT General Relativity Informal Tea-Time Series, Cambridge, MA, USA
- 05/2025 S250328ae: Joint observations with DECam and the Prime Focus Instrument at Subaru Observatory, The Dark Energy Survey Collaboration Meeting 2025, Virtual
- 05/2025* Status of the Rubin Target of Opportunity Program, OpenMMA Telecon, Virtual
- 03/2025 LSSTCam Performance and Readiness, APS March meeting 2025, Anaheim CA, USA
- 02/2025* Plenary: LSSTCam Performance and Readiness, DESC Collaboration meeting, Virtual
- 02/2025* LSSTCam Electro-optical Testing Status and Discussion, DESC Collaboration meeting, Virtual
- 11/2021* LLAMAS Assembly Integration and Testing, Wheaton College, Norton MA, USA
- 05/2020 Undergraduate honors thesis research, Wheaton College, MA, USA, Virtual
- 09/2018 REU research on dwarf satellite galaxies, Wheaton College, Norton MA, USA
- 08/2018 REU research on dwarf satellite galaxies, Rutgers University, New Brunswick NJ, USA

Posters

- 03/2025 LSSTCam Performance and Readiness, LSST@Europe7, Poznan, Poland
- 03/2025 LSSTCam Performance and Readiness, APS March meeting 2025, Anaheim CA, USA
- 03/2024 LSSTCam Defects, Image Sensors for Precision Astronomy 2024 at SLAC, Menlo Park CA, USA
- 06/2019 REU research on dwarf satellite galaxies, 234th AAS Meeting, St. Louis MO, USA
- 03/2019 Magnetic nanoparticle research, UCL Physics department poster symposium, London UK
- 08/2018 REU research on dwarf satellite galaxies, Rutgers University, New Brunswick NJ, USA
- 04/2018 Project P.A.N.O.P.T.E.S., Northeast Astronomy Forum, Suffern NY, USA